BEHIND THE SCREEN

Inside the Cloud: Data Centres and Physical Infrastructure

An educational guide for the Science, Technology & Society (STS) project

Central question
What physical systems make the seemingly invisible cloud work—and what happens when those systems fail?

Data-centre components • photo data journey • reliability • power and cooling • energy efficiency • real-world outage • quiz • research resources

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Use this guide from beginning to end, try the PUE calculation, and consult the sources for deeper research. Examples are illustrative unless explicitly identified as sourced facts.

1. Introduction: The cloud is physical

Cloud computing lets people use computing resources—storage, databases, software, and processing—over a network, often the internet. Although the word “cloud” can sound intangible, these services rely on physical infrastructure: buildings, computers, storage devices, network cables, power systems, cooling equipment, and people.

What is a data centre?

A data centre is a facility that houses computing and networking equipment used to store, process, and transmit digital information. Some support one organisation; others are operated by cloud providers and host services for many customers.

Why data centres matter

- Everyday services: photo storage, streaming, online games, messaging, banking, maps, and websites may depend on data-centre infrastructure.
- Business and public services: organisations use remote computing resources to run applications, store records, analyse information, and collaborate.
- Society: digital infrastructure affects privacy, security, availability, energy demand, land and water use, and the location of jobs and investment.

A simple analogy

Think of a data centre as a combination of a factory, library, and transport hub. Servers do the processing, storage systems keep information, and network equipment moves information. Power, cooling, security, and monitoring keep the whole operation working.

Main idea: The cloud is not a magical place in the sky. It is a service delivered by physical systems in real locations.

Questions to investigate

- Where is information stored when I upload a photo?
- What equipment is needed to make a website respond?
- How can a service stay available if a component fails?
- How much energy does a data centre use, and how can impact be reduced?

2. Inside a data centre: Main components

2.1 Servers — the workers

A server is a computer that provides a service or resource to other computers. Servers run applications, process requests, perform calculations, and communicate with storage and network systems. Important parts include the CPU, memory (RAM), local storage, and network interfaces. Example: when a user opens a website, an application server may process the request and retrieve information from a database.

2.2 Storage — where data is kept

Storage systems save information so it can be retrieved later. They may use hard disk drives (HDDs), solid-state drives (SSDs), or other technologies. Cloud services may divide data into objects or blocks, keep metadata to locate it, and use replication or other durability methods.

2.3 Network switches and routers — the junctions

A network switch connects devices within a network and forwards traffic to the appropriate destination. A router forwards packets between different IP networks. Together with cables and fibre links, they connect servers to storage systems and users.

2.4 Power systems — the lifeline

Power distribution equipment routes electricity to IT equipment. An uninterruptible power supply (UPS) can provide short-term backup power during an interruption. Generators may supply electricity for longer outages, depending on facility design. These systems need testing and maintenance.

2.5 Cooling — removing heat

Servers use electricity and produce heat. Cooling systems move heat away from IT equipment so temperatures remain within operating limits. Facilities may use fans and air conditioning, chilled-water systems, liquid cooling, or combinations of technologies.

2.6 Physical and digital security

Physical controls may include restricted access, identity checks, cameras, alarms, and monitoring. Cybersecurity controls protect accounts, networks, operating systems, applications, and data. Security is layered; no single measure is sufficient on its own.

How the parts work together

Component	Main job	Analogy
Server	Processes requests and runs applications	Worker
Storage	Saves and retrieves information	Library
Switch	Connects devices within a network	Local junction
Router	Moves traffic between networks	Road interchange
UPS / generator	Provides backup power	Emergency power
Cooling	Removes heat	Ventilation system

Real facilities differ. This is a simplified overview; large cloud facilities may contain multiple layers of servers, storage, networking, electrical distribution, and cooling.

3. Follow a photo: How cloud data moves

Imagine uploading a photo from a phone to a cloud app. The exact details vary by service, but the general journey often includes these stages:

Step 1 — Upload request

The app prepares the photo and sends a request over a network. The request may include account information, permissions, file name, and other metadata.

Step 2 — Network transport

Routers move packets between networks, while switches connect devices within networks. Data travels through wired or wireless connections and internet infrastructure.

Step 3 — Application server

The service receives the request and may authenticate the user, check permissions, validate the file, and decide where it should be stored.

Step 4 — Storage

The service writes the photo data to storage and records metadata so it can be located later. Some services use redundancy or multiple copies to improve durability.

Step 5 — Confirmation

After processing the upload, the service sends a response to the app. The app can then show that the upload succeeded.

Step 6 — Retrieve later

When the user opens the photo again, the service locates the stored data and sends it back to the device for display.

Important: Real services may use caches, content-delivery networks, file scanning, encryption, data partitioning, multiple data centres, and other components.

Simplified data-flow diagram

Phone / laptop

↓ Upload request through network

Internet → routers / switches

↓

Application server ↔ storage / database

↑ Confirmation or requested file returns to device

4. Reliability: What if something fails?

Data-centre engineers plan for component failures because computers, power equipment, network links, and cooling systems can all have faults. Reliability engineering tries to reduce both the chance and impact of failures.

Power interruption

Without protection, equipment may shut down. A UPS can bridge short interruptions, and a generator may support longer outages. Protection depends on design, maintenance, batteries, fuel, and correct transfer procedures.

Cooling failure

If cooling stops, temperatures may rise. Monitoring systems can alert staff; equipment may reduce performance or shut down to protect itself. Redundant cooling equipment and good airflow design can reduce risk.

Network link failure

A broken link may make equipment unreachable even if it is still running. Alternate network links, routing, and diverse paths can maintain connectivity when designed and configured correctly.

Storage drive failure

A failed drive may make some data unavailable. Systems can use replication, RAID, erasure coding, monitoring, and backups, depending on service requirements.

Redundancy vs backup

Redundancy means extra components or paths that help a service continue when one part fails. Example: a second network path.

Backup means a recoverable copy of data maintained so information can be restored after loss or corruption. Example: restoring a file from a previous backup.

Replication is not always the same as backup. Accidental deletion or corrupted data can be copied to a replica. Separate backups with suitable retention and regular recovery tests provide another layer of protection.

Common reliability principles

- Avoid single points of failure where consequences would be severe.
- Monitor equipment and alert staff when conditions become abnormal.
- Test backup power, recovery procedures, and backup restoration.
- Use controlled change processes and document incidents.
- Plan for shared dependencies: systems may still rely on the same power, network, or software component.

5. Energy, cooling and environmental impact

Why data centres use electricity

Electricity powers processors, memory, storage, network equipment, fans, pumps, lighting, and other facility systems. The IT equipment's electricity use ultimately becomes heat, which must be removed to keep equipment within operating limits.

Air cooling and liquid cooling

Air cooling: Fans move air through or around equipment. Cooling systems remove heat from the air and transfer it away from the server area.

Liquid cooling: A circulating coolant carries heat away, sometimes through cold plates attached to processors. Liquid cooling does not necessarily mean liquid touches electronic components directly.

PUE: Power Usage Effectiveness

PUE is a common data-centre energy-efficiency metric. It is calculated as:

PUE = Total data-centre facility energy ÷ IT equipment energy

Example: if a facility uses 1,000 kWh in total and IT equipment uses 700 kWh, then PUE = 1,000 ÷ 700 ≈ **1.43**. For every 1 kWh used by IT equipment, around 0.43 kWh is used by other facility systems in this example.

Try a calculation

- Total facility energy = 2,400 kWh
- IT equipment energy = 1,800 kWh
- PUE = 2,400 ÷ 1,800 = 1.33 (approximately)

What PUE does—and does not—tell us

- A lower PUE generally means less non-IT facility energy relative to IT energy.
- PUE does not directly measure useful computing output, carbon emissions, or water consumption.
- Environmental impact also depends on electricity sources, cooling choices, water availability, hardware efficiency, and how fully equipment is used.

Where to build a data centre

Site selection may consider reliable electricity, network connectivity, climate, water availability, land, local hazards, energy costs, and regulations. These factors interact; there is no single best location for every facility.

When using statistics, identify the geography and year. A report about U.S. electricity use should not be presented as a global or India-specific estimate.

6. Case study: The 2017 Amazon S3 outage

On 28 February 2017, an operational error disrupted Amazon Simple Storage Service (Amazon S3) in the Northern Virginia region. AWS published an explanation of the incident. This case study illustrates how a change in one subsystem can affect other services that depend on it.

1 — Investigation

The team was investigating an issue affecting the S3 billing system in the Northern Virginia region.

2 — Incorrect input

According to AWS's report, an authorised team member entered a command intended to remove a small number of servers, but an incorrect input caused more servers to be removed than intended.

3 — Wider impact

Some of the removed servers supported other S3 subsystems. The disruption affected S3 requests and other AWS services that depended on S3 in that region.

4 — Recovery

AWS restarted the affected subsystems and restored capacity. Recovery took time because systems had to be restarted and validated.

Lessons from the incident

- Human factors matter: operational errors can occur even when people are authorised.
- Dependencies matter: services often depend on shared infrastructure, so failures can spread.
- Guardrails matter: controlled changes, safer tooling, and checks can reduce unintended effects.
- Recovery takes time: restoring a large service may require restart, validation, and monitoring.

Accuracy note: This was an operational-capacity incident, not a cooling failure or a power outage. Use the original AWS report as the primary source.

Primary source: AWS — Summary of the Amazon S3 Service Disruption in the Northern Virginia (US-EAST-1) Region

8. Research resources and how to cite them

These sources provide background, technical explanations, primary-source incident information, and research on energy. Record the date you accessed each source.

AWS — What is a Data Center?
Introductory overview of data centres and cloud infrastructure.

Google Data Centers
Facility information, security, locations, and sustainability.

Cloudflare — What Is a Data Center?
Beginner-friendly explanation of data centres and internet services.

AWS — Regions and Availability Zones
Geographical infrastructure and fault isolation.

AWS — Official 2017 S3 disruption explanation
Primary-source case study about the operational error and service disruption.

Uptime Institute — Annual Outage Analysis 2025
Industry findings on outages and reliability; some details may require access.

Lawrence Berkeley National Laboratory — U.S. Data Center Energy Usage Report: 2025 Update
Research on energy use in the United States; not a global or India-specific estimate.

U.S. Department of Energy — Data Center Energy Efficiency Fact Sheet
Energy management and efficiency approaches.

How to use sources responsibly

- Prefer primary sources for incident details and official technical documentation for definitions.
- Use research institutions and industry reports for energy and reliability trends.
- Check publication date, geographic coverage, assumptions, and methodology.
- Do not present provider marketing statements as independent measurements.
- Cite factual claims that rely on a source, following the citation style required by your instructor.

Example reference format

Organisation. (Year or n.d.). *Title of page or report*. Website or publisher. Accessed 24 October 2026. URL

Use the actual publication date if available; “n.d.” means no date was stated. Adjust this format to your course's required style (APA, MLA, IEEE, or another style).